Forkful — Final POE Presentation Guide

PROG7314 POE

Step-by-Step Recording Script & Rubric Verification Walkthrough

Isabelle Devlin	Jamie Abrahams	Justin Fussell	Matteo Nusca
ST10445500 (Presenter)	ST10434047	ST10280758	ST10440432

TARGET DURATION	TARGET DEVICE	PRODUCTION BUILD	REPOSITORY TAG
7 – 10 Minutes	Physical Android Mobile	Forkful-Release.apk (Signed v2)	Final-POE (Main)

✓ Pre-Recording Checklist (2 Minutes)

- **Install Build:** Tap `Forkful-Release.apk` on your phone to install the production release.
- **Device Settings:** Ensure your phone has biometric security (fingerprint/face recognition) and a Google Account configured.
- **Recording Mode:** Screen record with microphone audio enabled, or mount a camera recording the physical device.
- **Second Screen Ready:** Have your laptop browser open to the Supabase dashboard (showing Edge Functions & Tables) for the 60-second backend section.

1. Introduction, Google SSO & Authentication Flow

Fixes Part 2 Feedback

- **Cold Launch & Zero-Flicker:** Open Forkful from your home screen. Point out that the theme loads instantaneously from encrypted cache with zero flash.
- **Google Single Sign-On:** Tap *"Continue with Google"*. The native Credential Manager prompt appears. Select your account to sign in smoothly.
- **Silent Token Refresh & Expiry Architecture:** Mention that auth tokens are encrypted via AES-256 and silently refreshed via HTTP 401 interceptors.
- **Sign Out & Instant Registration:** Go to Settings -> Log Out. Tap *Sign Up*, enter a test name, email, and password. Tap *Create Account*. Show that you are instantly authenticated with zero email confirmation delays or inbox barriers.
- **Returning User Onboarding Check:** Show that signing back in preserves onboarding preferences and routes straight to Explore without forcing re-onboarding.

> **SUGGESTED VOICEOVER:**
> *"Hi, I'm Isabelle Devlin presenting on behalf of Group 1. Today I'm demonstrating Forkful, our native Android culinary platform. We resolved all feedback from Part 2: Google SSO is fully integrated using Android Credential Manager with transparent token refresh, and new account sign-up auto-confirms instantly without any email barrier."*

2. Settings, Theming & South African Multi-Language

Core Accessibility

- **Dark & Light Theme Toggle:** Navigate to Settings. Toggle between Dark and Light mode, showcasing instant theme application.
- **Multi-Language Localization:** Tap *Language Selection*. Switch to **isiZulu (🇿🇦)** to demonstrate real-time localized string adaptation across tabs, then switch to **Afrikaans (🇿🇦)**, and back to **English**.
- **Biometric Security Lock:** Toggle *Biometric Lock* ON. Authenticate with your fingerprint/face prompt. Minimize the app to the Android home screen and reopen it—show that `AppLockScreen` immediately protects private kitchen data.

> **SUGGESTED VOICEOVER:**
> *"Forkful supports multi-language accessibility across official South African languages including isiZulu and Afrikaans. Native biometric security gates kitchen preferences upon cold start and whenever the app resumes from the background."*

3. Recipe Discovery Deck & Swipe Management

Fixes Part 2 Feedback

- **Swipe Deck Navigation:** Open the Explore tab. Show mood chips and recipe cards.
- **Left Swipe (Rejection Exclusion):** Swipe LEFT on a recipe card. Explain that swiped cards are permanently excluded from recurring in the active deck.
- **Right Swipe (Save & Grocery Queue):** Swipe RIGHT on a recipe card to bookmark it and automatically queue missing ingredients.
- **Unsave a Saved Recipe:** Open *Recipe Box*, select a saved recipe, and tap the heart icon. Show that it unsaves cleanly via the REST API.

> **SUGGESTED VOICEOVER:**
> *"In Part 3, we addressed the deck recursion issue: left-swiping excludes recipes from reappearing. Furthermore, users can now cleanly unsave any recipe by tapping the heart toggle in the Recipe Details view."*

4. Smart Pantry & Strict Matching Algorithm

Fixes Part 2 Feedback

- **Pantry Initial Load:** Tap the Pantry tab. Show the initial loading indicator resolving smoothly without flashing an empty state.
- **Edit Existing Pantry Items:** Tap the *Edit* icon on an existing item. Modify the quantity (e.g., from 500g to 1kg) and update the expiry date slider. Tap *Save* and show the updated entry.
- **Strict Whole-Word Matching:** Point out the Smart Pantry Matches section. Explain that the algorithm now enforces strict whole-word tokenized matching so 'salt' never incorrectly matches 'salted butter'.

> **SUGGESTED VOICEOVER:**
> *"We unified our pantry matching rules across all endpoints. Unlike Part 2's loose substring matching, our tokenized engine prevents false positives like salt matching salted butter. Users can also edit any existing ingredient's quantity and shelf life."*

5. Resilient Offline Mode & Automatic Synchronisation

RoomDB & SQLite Architecture

- **Simulate Network Disconnection:** Pull down Android Quick Settings and turn on **Airplane Mode**.
- **Visual Offline Feedback:** Point out the alert banner: *"Offline - viewing cached pantry"*.
- **Optimistic Local Write:** Tap + to add an ingredient (e.g., *"Fresh Garlic"*). Show it immediately stored in RoomDB with an offline pending sync badge.
- **Reconnection & Automatic Sync:** Turn Airplane Mode OFF. Watch the status banner automatically reconcile with the cloud and update to *"Synced at HH:mm"*.

> **SUGGESTED VOICEOVER:**
> *"Forkful delivers full offline resilience using Android RoomDB. Even without cellular or Wi-Fi connectivity, users can manage stock offline. As soon as the connection returns, our background engine flushes pending writes to Supabase automatically."*

6. Aisle-Organised Basket & Working Actions Menu

Fixes Part 2 Feedback

- **Supermarket Aisle Grouping:** Open the Basket tab. Point out items categorized under aisles (Produce, Dairy, Spices, Bakery).
- **Add Items by Hand:** Tap the + icon in the header to add an item manually (e.g., *"Olive Oil"*).
- **Delete Items:** Tap the trash icon on an item to remove it individually.
- **Functional Basket Menu (⋮):** Tap the ⋮ menu button. Demonstrate *"Share basket"* opening the native Android share sheet for WhatsApp, and *"Remove collected items"* clearing checked entries.

> **SUGGESTED VOICEOVER:**
> *"Addressing Part 2 feedback, the grocery basket now allows manual item addition, individual deletion, and a fully functional options menu with WhatsApp export and bulk clearing."*

7. Real-Time Push Notifications (Firebase FCM)

Cloud Messaging

- **Notifications Screen:** Tap the notification bell icon.
- **Simulate Expiry Alert:** Tap *"Simulate Expiry Alert"* to show a high-priority heads-up notification in the Android system tray.
- **Simulate Meal Suggestion:** Tap *"Simulate Meal Recommendation"* to demonstrate contextual meal reminders based on current pantry ingredients.

8. Backend & Cloud Infrastructure Walkthrough (60 Seconds)

Fixes Part 2 Feedback

- **Camera on Laptop Screen:** Point the camera at the Supabase project dashboard.
- **Edge Functions:** Show active deployed functions: `auth-sso`, `auth-refresh`, `recipes-deck`, `pantry`, and `pantry-match`.
- **Relational Database:** Briefly show PostgreSQL tables: `recipes`, `user_pantry`, `user_swipes`, and `user_preferences` with Row Level Security (RLS) policies.
- **CI/CD & Conclusion:** Mention the GitHub Actions automated CI pipeline, 100% passing unit tests, and production release tag `Final-POE`.

> **SUGGESTED VOICEOVER:**
> *"Here is our cloud backend on Supabase. We have 10 deployed Deno Edge Functions handling auth, recipe ranking, and inventory sync, backed by PostgreSQL with Row Level Security. All commits are pushed to main and tagged with Final-POE. Thank you!"*